Power Series Solutions to Nonlinear Ordinary Differential Equations
MATH 689
Homework 2

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1. HW1 Problem 2 revisited (adapted from problem 2.11 on p. 46)

$$f' + f = e^{x^3}, \quad f(0) = 1, \quad x \in [0,1] \tag{1}$$

(1a) In preparation for finding the power series solution of the above ODE, note that

$$e^{x^3} = \sum_{n=0}^{\infty} \frac{1}{n!}(x^3)^n = \sum_{n=0}^{\infty} b_n x^n, \quad b_n = \frac{1 + 2\cos\left(\frac{2n\pi}{3}\right)}{3(n/3)!} \tag{2}$$

1a.i. Confirm that these two series are equal as written above. You are only required to check up to the x^6 term for this.

We check equivalence of the two series up to the x^6 term by reducing the upper bound of the sum as

$$\begin{gathered} \sum_{n=0}^{2} \frac{1}{n!}(x^3)^n = 1 + x^3 + \frac{x^6}{2} \\ \sum_{n=0}^{6} \frac{1 + 2\cos\left(\frac{2n\pi}{3}\right)}{3(n/3)!} x^n = \sum_{n \in \mathcal{F}^{-1}(A)} \frac{1 + 2\cos\left(\frac{2n\pi}{3}\right)}{3(n/3)!} x^n \quad : \quad \mathcal{F}(n) = (n/3)! \text{ and } A = \{0,1,2,3,4,5,6\} \end{gathered} \tag{3}$$

In other words we only sum over the values of n that is well defined in the preimage of $\mathcal{F}$.

$$\begin{aligned} \sum_{n \in \mathcal{F}^{-1}(A)} b_n x^n &= \frac{x^0(1 + 2\cos(0))}{3(0/3)!} + \frac{x^3(1 + 2\cos(2\pi))}{3(3/3)!} + \frac{x^6(1 + 2\cos(4\pi))}{3(6/3)!} \\ &= 1 + x^3 + \frac{x^6}{2} \end{aligned} \tag{4}$$

1a.ii. Using the Taylor expansion formula, verify that the Taylor expansion of e^{x^3} about $x = 0$ agrees with the above. You are only required to check up to the x^3 term for this.

Recall that the n th coefficient of the Taylor expansion centered at $x = a$ is given by

$$a_n = \frac{1}{n!} \frac{d^n f(x)}{dx}\bigg|_{x=a} \tag{5}$$

We compute the following first 4th terms

$$\begin{aligned} 0!a_0 &= e^{0^3} = 1 \\ 1!a_1 &= 3(0)^2 e^{0^3} = 0 \\ 2!a_2 &= 6(0)e^{0^3} + 9(0)^4 e^{0^4} = 0 \\ 3!a_3 &= 6e^{0^3} + 18(0)^3 e^{0^3} + 36(0)^3 e^{0^3} + 27(0)^6 e^{0^3} = 6 \end{aligned} \tag{6}$$

it is clear from the coefficients we have agreement with the above in (4) up to the x^3 term .

1b. Find the power series solution of the ODE. That is, assume $f = \sum_{n=0}^{\infty} a_n x^n$ and determine a recurrence relation for the a_n's. You may write your answer in terms of the b_n coefficients determined above (i.e., you can just write b_n in the recursion and write the formula for b_n off to the side).

We assume that f has a solution in the form of a power series, where $f = \sum_{n=0}^{\infty} a_n x^n$, then from (1) we have

$$f' + f = \sum_{n=0}^{\infty} a_{n+1}(n+1)x^n + \sum_{n=0}^{\infty} a_n x^n = e^{x^3} = \sum_{n=0}^{\infty} b_n x^n \tag{7}$$

Equating the sequence of coefficients in (7) we obtain

$$a_{n+1} = \frac{b_n - a_n}{n+1}, \quad a_0 = 1, \quad a_1 = 1 \qquad b_n = \frac{1 + 2\cos\left(\frac{2n\pi}{3}\right)}{3(n/3)!} \tag{8}$$

We include $a_1 = 1$ since Matlab starts indexing at 1.

1d. Plot successive truncations of the series on the same graph as the numerical solution. Choose a large enough N value such that series has converged to the extent that it is indistinguishable (to the naked eye) with the numerical solution over the full computational domain.

See in (1c) we choose $N = 25$, and plot every 5th adtional term to the series. It is apparent that from our plot the 10 th term and onward guarantees that the series solution has converged to the extent that it is indistinguishable.

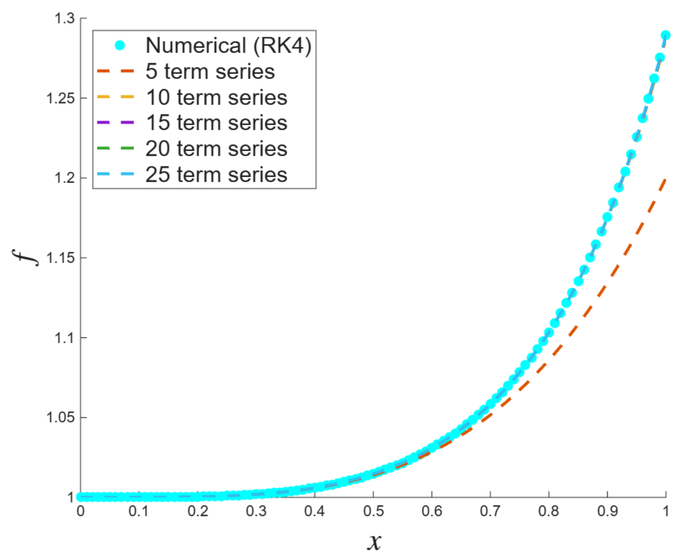

Figure 1: Power Series and Numerical Solution to (1)

I put 1d before 1c for space organization reasons. 1c is more chucky.

1c. Incorporate the power series solution into the code you wrote for HW1 Problem 2. A template is provided in lines 30-44 of Example1withSeries.m.

MATLAB - RK4 and Power Series Solution

```
clear;clc;clf
%%Class Example 1 on 1/16/2024
dx=0.00001; %STEP SIZE
x=0:dx:1; %DOMAIN OF INDEPENDENT VARIABLE
f=x; %PREALLOCATE MEMORY FOR DEPENDENT VARIABLE (will be overwritten below)
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=1; %this seems silly now but is a placeholder for higher-order ODEs
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)) ;
    k3=F(u+k2*dx/2,x(n)) ; k4=F(u+k3*dx,x(n)) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',

Color="cyan");
hold on
%POWER SERIES COEFFICIENTS
N=25; %max number of terms to take in the series
a0=1; %coefficient(s) that come(s) from initial condition(s)
a(1)=0; %coefficient(s) that involve n=0 or are obtained non-recursively
for n=1:N
    b(n)=(1+2*cos(2*n*pi/3))/3/gamma((n+3)/3);
    a(n+1)=(b(n) - a(n))/ (n+1) ;
end
fS=a0; axis1=axis; %this saves the nice axis range provided by the numerical solution
for n=1:N
    fS=fS+a(n)*x.^n;
    if round(n/5) == n/5
       plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end; axis(axis1)
end
legend('location','northwest','fontsize',15)
legend show;
xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)
%1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x); f=u;
dudx=[exp(x.^3)-u];
end
```

2. HW1 Problem 3 revisited (adapted from problem 2.9b on p. 46)

$$f'' + f = 0, \quad f(0) = 0, \quad f'(0) = 1, \quad x \in [0, 10] \tag{9}$$

2a. Find the power series solution. That is, assume $f = \sum_{n=0}^{\infty} a_n x^n$ and determine a recurrence relation for the a_n's.

We first derive the first and second derivatives of f assuming it takes the form of a power series.

$$f(x) = \sum_{n=0}^{\infty} a_n x^n, \quad f'(x) = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n, \quad f''(x) = \sum_{n=0}^{\infty} (n+2)(n+1) a_{n+2} x^n \tag{10}$$

We plug into (9) and adjusted index shifts accordingly

$$f'' + f = \sum_{n=0}^{\infty} \left[(n+2)(n+1) a_{n+2} + a_n \right] x^n = 0 \tag{11}$$

The natural next thing to redefine our initial conditions in terms of coefficients,

$$f(0) = a_0 + \sum_{n=1}^{\infty} a_n 0^n = 0, \quad f'(0) = a_1 + \sum_{n=1}^{\infty} (n+1) a_{n+1} 0^n = 1 \tag{12}$$

We then solve for a_{n+2} from (11)

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)} \tag{13}$$

Using our initial values a_0, a_1 we look at the first few terms of the recurrence relation (13) .

$$a_2 = 0, \quad a_3 = -\frac{1}{3!}, \quad a_4 = 0, \quad a_5 = \frac{1}{5!}, \quad a_6 = 0, \quad a_7 = -\frac{1}{7!} \tag{14}$$

It is apparent that for even and odd $n's$ we have

$$a_{2n} = 0, \quad a_{2n+1} = \frac{(-1)^n}{(2n+1)!} \tag{15}$$

But summing over infinite zeros is still zero thus we eliminate the a_{2n} terms from the sum

$$f(x) = \sum_{n=0}^{\infty} a_{2n+1} x^{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = \sin(x) \tag{16}$$

2b. Verify that the power series solution obtained above is the same as the Taylor expansion about $x = 0$ of the analytical solution you found in HW1 (problem 3d). Confirm this for the first 5 non-zero terms of the series/expansion.

WLOG, we can say that the analytical solution $\sin(x)$ is defined to be the power series that we have notice that is equivalent to the analytical solution in (16). However we can still check for the sake of the problem. We have the partial sum of the power series solution as

$$\sum_{n=0}^{4} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} \tag{17}$$

Using (5) on $\sin(x)$ about $x = 0$ we have

$$\begin{aligned}
0! a_0 &= \sin(0) = 0, \quad 1! a_1 = \cos(0) = 1 \\
2! a_2 &= -\sin(0) = 0, \quad 3! a_3 = -\cos(0) = -1 \\
4! a_4 &= \sin(0) = 0, \quad 5! a_5 = \cos(0) = 1 \\
6! a_6 &= -\sin(0) = 0, \quad 7! a_7 = -\cos(0) = -1 \\
8! a_8 &= \sin(0) = 0, \quad 9! a_9 = \cos(0) = 1
\end{aligned} \tag{18}$$

For non-zero coefficients on the left side we, if we divide the factorial coefficient over we we see that this is agreement with coefficients of the powerseries solution in (17).

2c. Incorporate the power series solution into the code you wrote for HW1 Problem 3.

MATLAB

```
close all
clear;clc;clf
%%Class Example 1 on 1/16/2024
dx=0.0001; %STEP SIZE
x=0:dx:10; %DOMAIN OF INDEPENDENT VARIABLE
f=x; %PREALLOCATE MEMORY FOR DEPENDENT VARIABLE (will be overwritten below)
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=[0 1]; %this seems silly now but is a placeholder for higher-order ODEs
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)) ;
    k3=F(u+k2*dx/2,x(n)) ; k4=F(u+k3*dx,x(n)) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4); f(n+1)=u(1);
end; hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color="cyan");
hold on
%POWER SERIES COEFFICIENTS
N=100; %max number of terms to take in the series
a0=0; %coefficient(s) that come(s) from initial condition(s)
a(1)=1;
a(2)=0;
%coefficient(s) that involve n=0 or are obtained non-recursively
for n=1:N
    a(n+2)=-a(n) / (n+2)/(n+1);
end
fS=a0; axis1=axis; %this saves the nice axis range provided by the numerical solution
for n=1:N
    fS=fS+a(n)*x.^n;
    if round(n/20) == n/20
       plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end; axis(axis1)
end
fA=sin(x); % use .* and .^ and ./
plot(x,fA,'LineWidth',2,'DisplayName','Analytical Solution')
legend('location','northwest','fontsize',10)
legend show;
xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)
%1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x)
f=u(1); v=u(2);
dudx=[v -f];
end
```

2d. Plot successive truncations of the series on the same graph as the numerical and analytical solutions. Choose a large enough N value such that series has converged to the extent that it is indistinguishable (to the naked eye) with the numerical and analytical solutions over the full computational domain.

In this plot we choose $N = 100$, and plot every 20th additional term in the series. It seems that beyond 40th term we have near guaranteed convergence.

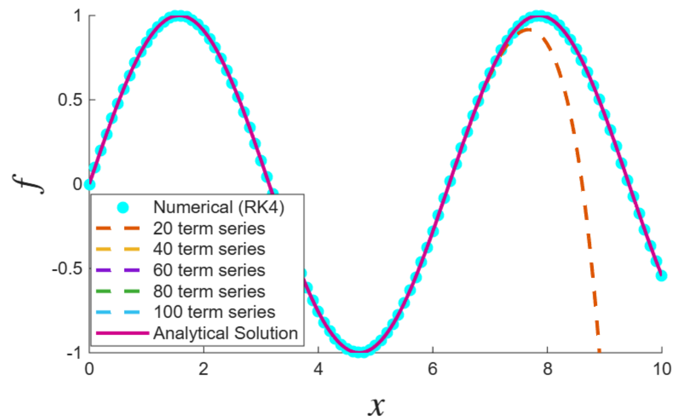

Figure 2: Numerical and Power Series Solution to (9)

3. Adapted from problem 2.14 on p. 47:

The following is known as a Weber equation. Its solution is its own special function.

$$f'' + \frac{1}{4}x^2 f = 0, \quad f(0) = 1, \quad f'(0) = 0, \quad x \in [0, 4] \tag{19}$$

3a. Modify one of your existing codes (such as ClassExample2.m) to solve the above problem numerically over the domain specified above.

MATLAB

```
dx=0.0001; %STEP SIZE
x=0:dx:4; %DOMAIN OF INDEPENDENT VARIABLE
f=x;
%INITIAL CONDITION VECTOR (or value, if ODE is 1st-order)
u0=[1,0];
%NUMERICAL SOLUTION (RK4 ENGINE)
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)) ;
    k3=F(u+k2*dx/2,x(n)) ; k4=F(u+k3*dx,x(n)) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4);  f(n+1)=u(1);
end; figure(1); hold on
%PLOT NUMERICAL SOLUTION, DOWNSAMPLE POINTS
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)');
%POWER SERIES COEFFICIENTS
N=100; %max number of terms to take in the series
a0=1; a(1)=0;%coefficient(s) that come(s) from initial condition(s)
a(2)=0; a(3)=0; a(4)= -1/48;  %coefficient(s) that involve n=0 or are obtained non-recursively
for n=3:N %get all the rest of the coefficients
    a(n+2)=-a(n-2)/4/(n+1)/(n+2);
end
%PLOT THE POWER SERIES
fS=a0; axis1=axis; %this saves the nice axis range provided by the numerical solution
for n=1:N
    fS=fS+a(n)*x.^n;
    if round(n/10)==n/10  %this skips every 5 terms
        plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end; axis(axis1)
end
legend('location','northeast','fontsize',10)
xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)
% 1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x)
f=u(1); v=u(2);
dudx=[v -x^2*f/4];
end
```

3b. Find the power series solution. That is, assume $f = \sum_{n=0}^{\infty} a_n x^n$ and determine a_2, a_3, and a recurrence relation for the a_n's for $n > 3$.

$$\begin{aligned} 0 = f'' + \frac{x^2}{4} f &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \frac{1}{4}x^2 \sum_{n=0}^{\infty} a_n x^n \\ &= \sum_{n=0}^{\infty} (n+2)(n+1)a_{n+2}x^n + \frac{1}{4} \sum_{n=0}^{\infty} a_n x^{n+2} \\ &= 2a_2 + 6a_3 x + \sum_{n=2}^{\infty} (n+2)(n+1)a_{n+2}x^n + \frac{1}{4} \sum_{n=2}^{\infty} a_{n-2}x^n \\ &= \sum_{n=2}^{\infty} \left[(n+2)(n+1)a_{n+2} + \frac{1}{4} a_{n-2} \right] x^n \end{aligned} \tag{20}$$

We set $2a_2 = 0$ and $6a_3 = 0$ because if we match with the 0 ("forcing term") we should get that all possible coefficients of the polynomial is 0. Then we have the recurrence relation as

$$a_{n+2} = \frac{-a_{n-2}}{4(n+2)(n+1)}, \quad \text{for } n > 3. \tag{21}$$

With $a_0 = 1, a_1 = 0, a_2 = 0, a_3 = 0, a_4 = -1/48, a_5 = 0$.

3c. Incorporate the power series solution into the code. *shown in 3a and 3d.*

3d. Plot successive truncations of the series on the same graph as the numerical solution. Choose a large enough N value such that series has converged to the extent that it is indistinguishable (to the naked eye) with the numerical solution over the full computational domain.

We choose $N = 100$ and plot every 10th additional term of the series, and notice that beyond 10th term guarantees convergence to the solution on the interval $x \in [0, 4]$.

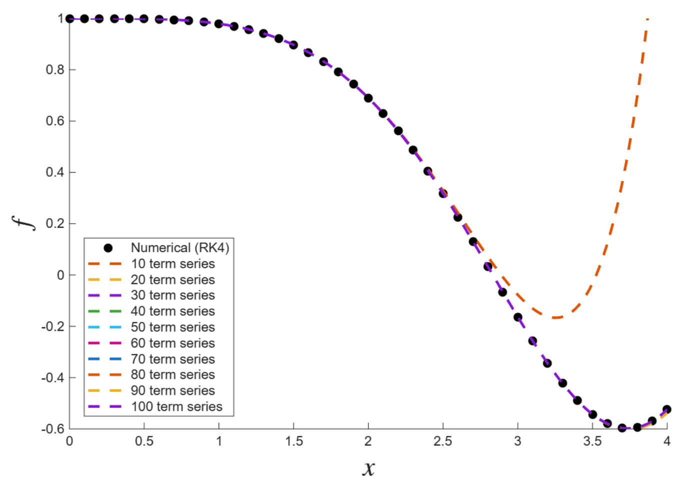

Figure 3: Power Series and RK4 Solution to (19)

4. MATH 689 Problem, adapted from problem 2.12 on p. 47:

$$f' + f = \frac{1}{1+x}, \quad f(0) = 1, \quad x \in [0, 2] \tag{22}$$

4a. Modify one of your existing codes to solve the above problem numerically over the domain specified above. *Answer on page 10.*

4b. In preparation for finding the power series solution of the above ODE, note that

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n \tag{23}$$

Confirm that the Taylor expansion of $\frac{1}{1+x}$ about $x = 0$ agrees with the above (you are only required to check up to the x^3 term for this). We take the truncated sum as

$$\sum_{n=0}^{3} (-1)^n x^n = 1 - x + x^2 - x^3 \tag{24}$$

Then we find the coefficients to to the Taylor expansion of $\frac{1}{1+x}$ about $x = 0$ up to the x^3 term using (5),

$$\begin{aligned} 0!a_0 &= \frac{1}{1+0} = 1, \quad 1!a_1 = \frac{-1}{(1+0)^2} = -1 \\ 2!a_2 &= \frac{-2}{(1+0)^3} = 2, \quad 3!a_3 = \frac{-6}{(1+0)^4} = -6 \end{aligned} \tag{25}$$

Again, this agrees with the coefficient in (24).

4c. Find the power series solution of the ODE. That is, assume $f = \sum_{n=0}^{\infty} a_n x^n$ and determine a recurrence relation for the a_n's.

$$\begin{aligned} \sum_{n=0}^{\infty} a_{n+1}(n+1)x^n + \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (-1)^n x^n \\ \sum_{n=0}^{\infty} x^n \left[a_{n+1}(n+1) + a_n\right] &= \sum_{n=0}^{\infty} (-1)^n x^n \end{aligned} \tag{26}$$

Thus we have the recurrence relation as

$$a_{n+1} = \frac{(-1)^n - a_n}{n+1} \tag{27}$$

with $a_0 = 1, a_1 = 0$, using notation from class we have a recurrence of $a_{n>1}$.

4d. Incorporate the power series solution into the code. *Answer on page 10.*

4e. Plot successive truncations of the series on the same graph as the numerical solution. At approximately what x value (and beyond) does the series fail to agree with the exact solution, no matter how many terms you include in the series?

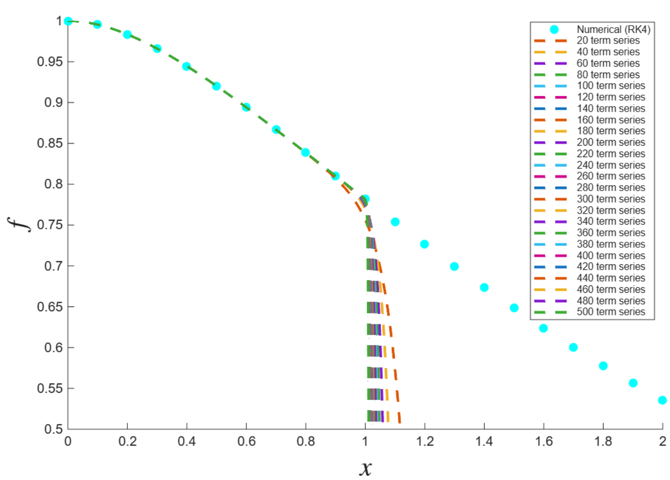

Figure 4: Power Series Solutions on $x \in [0, 1)$ and RK4 Solution

Recall from MATH 431, we have the radius of convergence as

$$R = \left(\left| \limsup_{n} \frac{Q_{n+1}}{Q_n} \right| \right)^{-1} \tag{28}$$

Where $Q_{n+1} = \frac{(-1)^n - a_n}{n+1}$ in our solution. We will not prove the full thing here but essentially

$$\limsup_{n\to\infty} \left| \left(\frac{n}{n+1} \left(\frac{(-1)^n - a_n}{(-1)^{n-1} - a_{n-1}} \right) \right) \right| \approx \left| \limsup_{n\to\infty} \left(\frac{n}{n+1} \right) \right| = 1 \tag{29}$$

It is clear that we can say both $|a_n|$ and $|a_{n-1}|$ behaviors similarly in the long run. And also note that taking the absolute value of both the $(-1)^n$ will go to 1 so we are left with the $\frac{n}{n+1}$ term. We accept that the power series only give exact solution on $x \in (-1, 1)$, due to the singularity at $x = -1$. Then because of the radius of convergence of 1, the power series solutions for (22) is only defined on $x \in (-1, 1)$.

Answer to both 4a and 4d, with graph in 4e.

MATLAB

```
dx=0.0001;
x=0:dx:2;
f=x;
u0=1;
f(1)=u0(1); u=u0;
for n=1:length(x)-1
    k1=F(u,x(n)) ; k2=F(u+k1*dx/2,x(n)) ;
    k3=F(u+k2*dx/2,x(n)) ; k4=F(u+k3*dx,x(n)) ;
    u=u+(dx/6)*(k1+2*k2+2*k3+k4);  f(n+1)=u(1);
end; figure(1); hold on
plot(x(1:1e3:end),f(1:1e3:end),'k.','markersize',20,'displayname','Numerical (RK4)',
Color="cyan");
hold on
N=500;
a0=1;
a(1)=0;
for n=1:N
    a(n+1)=((-1)^n - a(n) )/ (n+1);
end
fS=a0; axis1=axis;
for n=1:N
    fS=fS+a(n)*x.^n;
    if round(n/20) == n/20
       plot(x,fS,'--','LineWidth',2,'DisplayName',[num2str(n) ' term series']);
    end; axis(axis1)
end
legend('location','northwest','fontsize',7.5)
legend show;
xlabel('$x$','Interpreter','latex','FontSize',20)
ylabel('$f$','Interpreter','latex','FontSize',20)
%1st order ODE SYSTEM (or single equation, if 1st order already)
function dudx = F(u,x)
f=u(1); f=u;
dudx=[1/(1+x)-u] ;
end
```

Reference

[1] J. C. P. Miller, *Tables of Weber parabolic cylinder functions*, Her Majesty's Stationary Office, London, 1955.